

KILL DEVIL HILLS, NC, USA

WMO: 720282

Lat: 36.018N

Lon: 75.671W

Elev: 4

StdP: 101.28

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03736

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.3	(c) -1.4	(d) -14.8	(e) 1.0	(f) -2.4	(g) -11.3	(h) 1.4	(i) 1.4	(j) 8.9	(k) 5.0	(l) 8.2	(m) 4.9	(n) 4.0	(o) 310	(p) 0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.6	(c) 33.6	(d) 25.7	(e) 32.2	(f) 25.2	(g) 30.9	(h) 24.5	(i) 27.1	(j) 31.2	(k) 26.2	(l) 30.1	(m) 25.6	(n) 29.1	(o) 2.9	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 26.1	(b) 21.5	(c) 29.3	(d) 25.2	(e) 20.3	(f) 28.0	(g) 24.5	(h) 19.5	(i) 27.6	(j) 85.0	(k) 31.1	(l) 81.3	(m) 30.2	(n) 78.4	(o) 29.0	(p) 29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.9	(b) 7.0	(c) 5.8	(e) -5.8	(f) 36.2	(g) 2.7	(h) 1.7	(i) -7.7	(j) 37.4	(k) -9.3	(l) 38.4	(m) -10.8	(n) 39.3	(o) -12.8	(p) 40.5		
(4) 7.9	(5) 7.0	(6) 5.8	(7) -5.8	(8) 36.2	(9) 2.7	(10) 1.7	(11) -7.7	(12) 37.4	(13) -9.3	(14) 38.4	(15) -10.8	(16) 39.3	(17) -12.8	(18) 40.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	6.9	7.1	10.3	15.4	20.2	24.9	26.8	26.3	24.1	19.1	12.9	9.7	(6)
(7)		DBStd	8.23	5.15	4.70	5.05	4.31	3.84	2.91	2.23	2.29	2.53	3.97	4.26	4.77	(7)
(8)		HDD10.0	379	124	104	61	6	0	0	0	0	0	1	19	65	(8)
(9)		HDD18.3	1531	354	314	252	107	27	1	0	0	1	39	167	269	(9)
(10)		CDD10.0	2949	28	23	71	169	316	447	521	507	423	283	107	55	(10)
(11)		CDD18.3	1059	0	0	4	21	84	197	262	248	173	63	5	2	(11)
(12)		CDH23.3	8051	0	1	13	86	438	1556	2522	2186	1016	224	8	2	(12)
(13)		CDH26.7	2312	0	0	1	10	88	476	844	669	199	24	1	0	(13)
(14)	Wind	WSAvg	2.7	3.0	3.1	3.1	3.0	2.8	2.5	2.2	2.0	2.6	2.5	2.7	2.6	(14)
(15)	Precipitation	PrecAvg	1404	109	93	111	95	110	128	158	153	144	106	90	106	(15)
(16)		PrecMax	1872	191	197	224	178	204	239	254	291	506	222	235	218	(16)
(17)		PrecMin	979	45	36	32	34	42	46	93	60	36	3	21	43	(17)
(18)		PrecStd	187	36	41	38	35	42	47	38	54	98	49	52	39	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.9	21.5	25.2	28.0	31.0	35.0	35.6	36.0	32.4	28.9	24.1	22.4	(19)
(20)			MCWB	16.3	16.1	17.6	19.0	22.1	24.8	26.5	27.9	25.1	22.9	19.7	19.6	(20)
(21)		2%	DB	18.4	19.1	22.5	25.8	29.0	32.5	33.6	32.9	30.2	27.2	22.1	20.2	(21)
(22)			MCWB	16.1	14.8	16.1	18.7	21.9	24.4	25.9	26.1	24.3	22.7	18.4	18.2	(22)
(23)		5%	DB	16.9	17.1	20.1	23.8	27.4	31.0	32.3	31.4	28.8	25.8	20.3	18.5	(23)
(24)			MCWB	14.4	13.8	15.6	18.1	21.0	23.9	25.3	25.4	23.8	21.9	17.7	16.8	(24)
(25)		10%	DB	14.8	14.7	18.0	22.1	25.8	29.4	30.8	30.0	27.6	24.4	18.9	17.1	(25)
(26)			MCWB	12.6	11.8	14.9	17.4	20.6	23.3	24.7	24.9	23.3	21.4	17.0	15.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.2	17.8	19.0	21.7	24.0	26.4	27.9	28.9	26.8	24.7	21.0	20.5	(27)
(28)			MCD B	18.8	20.1	23.1	24.5	27.4	31.3	33.0	34.0	29.6	26.6	22.6	21.5	(28)
(29)		2%	WB	16.6	16.1	17.8	20.3	23.0	25.5	26.8	27.6	25.6	23.8	19.5	18.6	(29)
(30)			MCD B	18.3	18.1	20.7	23.7	26.8	30.0	31.9	31.4	28.2	25.8	21.0	20.0	(30)
(31)		5%	WB	14.9	14.1	16.7	19.3	22.3	24.9	26.1	26.5	24.9	23.0	18.3	17.1	(31)
(32)			MCD B	16.6	16.6	19.3	22.6	25.8	29.2	30.7	29.6	27.5	25.1	19.8	18.2	(32)
(33)		10%	WB	12.8	11.9	15.3	18.2	21.6	24.1	25.5	25.5	24.3	22.1	17.0	15.5	(33)
(34)			MCD B	14.6	14.3	17.9	21.5	24.8	28.2	29.5	28.6	26.9	24.2	18.7	16.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.9	7.3	7.5	7.8	7.2	6.8	6.6	6.0	5.2	5.9	6.2	6.4	(35)
(36)			MCDBR	9.5	11.2	10.9	10.6	9.4	8.8	8.6	7.9	6.8	6.9	7.7	8.2	(36)
(37)			MCWBR	7.9	8.2	6.7	5.5	4.4	3.7	3.7	3.3	3.3	3.6	5.6	6.9	(37)
(38)		5% WB	MCDBR	9.3	10.9	10.0	9.6	8.0	8.0	8.1	7.8	6.0	5.8	6.8	8.0	(38)
(39)			MCWBR	8.5	9.2	7.3	5.9	4.2	3.7	3.8	3.5	3.5	3.6	6.1	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.323	0.331	0.354	0.398	0.441	0.486	0.513	0.501	0.418	0.368	0.336	0.328	(40)	
(41)		taud	2.536	2.510	2.469	2.356	2.274	2.199	2.131	2.154	2.388	2.512	2.546	2.549	(41)	
(42)		Ebn at Noon	877	911	915	886	849	808	782	783	838	858	854	847	(42)	
(43)		Edh at Noon	80	91	104	122	135	145	154	148	112	90	78	75	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.36	3.15	4.23	5.40	6.08	6.44	6.08	5.53	4.58	3.67	2.67	2.13	(44)	
(45)		RadStd	0.17	0.32	0.32	0.39	0.44	0.35	0.31	0.37	0.29	0.36	0.19	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+159	+109

Nomenclature: See separate page